



RS004 Week 10 — English Worksheet

Topic: Paper to Code (Apply) · **Course:** Platformer Game · **Time:** about 45 minutes

This week you think about how a level is designed: how coordinates place each platform, and how jump height and jump distance decide what is reachable. You will read level code, find a design mistake on paper, and explain the code you wrote in your live lesson.

Keep these words handy: **sketch**, **coordinate (x, y)**, **jump height**, **jump distance**, **reachable**.

1 · Predict 🧠

Read each snippet and write what you think it means.

```
# JUMP_STRENGTH = -14, GRAVITY = 0.7
# max jump height  $\approx 14^2 / (2 \times 0.7) = 140$  pixels
```

If your player is on a platform, roughly how high can the next platform be and still be reachable?


```
MY_LEVEL = [
    (0, 580, 250, 20),
    (350, 520, 100, 20),
]
```

The first platform ends at $x = 250$. The next starts at $x = 350$. How wide is the gap to jump?


```
START_X, START_Y = 50, 540  
FINISH_X, FINISH_Y = 450, 220
```

Where does the player begin, and where is the goal? Lower **y** is higher on screen.

2 · Spot the Bug 🐛

Each snippet is a *level design* mistake. Read what's wrong, then rewrite the coordinates so the level works. Explain your fix.

Bug A — The second platform is meant to be reachable, but it is 300 pixels above the first — too high to jump.

```
MY_LEVEL = [  
    (0, 580, 250, 20),  
    (300, 280, 100, 20),  
]
```

Hint: max jump height is about 140 pixels. Lower the gap.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The finish marker is floating in empty space with no platform under or near it, so the player can never reach it.

```
MY_LEVEL = [  
    (0, 580, 250, 20),  
    (350, 520, 100, 20),  
]  
FINISH = (700, 100, 50, 40)
```

Hint: add a platform near the finish, within jump range of the last one.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

This is a working level. Each platform is `(x, y, width, height)`. The player starts low and climbs to the finish at the top.

```
START_X, START_Y = 50, 540

MY_LEVEL = [
    (0, 580, 250, 20),
    (350, 520, 100, 20),
    (520, 430, 100, 20),
    (350, 330, 100, 20),
]

FINISH = (380, 280, 50, 40)
```

What do the first two numbers in each platform tuple mean?

The gap from the first platform (ends at x = 250) to the second (starts at x = 350) is 100 pixels. Is that within jump distance? How do you know?

The player must climb from y = 580 up to the finish at y = 280. Why is a higher platform written with a *smaller* y number?

Look at the last platform (350, 330, 100, 20) and the finish (380, 280, 50, 40) . Why is the finish placed there and not somewhere far away?

If you wanted to make this level harder, which one number would you change, and why?

4 · Explain Your Lesson Code 🎥

Explain the level **you wrote in today's live lesson** in a short phone video. You may show it running. Try to use these words: **sketch, coordinate, jump height, reachable, difficulty**.

1. Show your code and point at your **MY_LEVEL** platforms.
2. Read the coordinates of one platform out loud and say where it sits.
3. Show the hardest jump and explain why it is still reachable.
4. Say one thing you would change to adjust the difficulty.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.